Sentiment Analysis of Airline Tweets

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Problem Statement:

Airline companies receive vast amounts of user feedback on Twitter, often expressing dissatisfaction or satisfaction. The objective of this project is to build a sentiment classification model that categorizes airline tweets as *positive*, *neutral*, or *negative* using Natural Language Processing (NLP) and machine learning techniques. With the growing impact of social media on brand reputation, understanding customer sentiment is crucial for maintaining service quality and customer trust.

Summary:

In this project, a sentiment analysis model was developed to classify airline tweets into three categories: positive, neutral, and negative. The dataset was preprocessed extensively to remove noise, and various NLP techniques were applied for vectorization. Several machine learning models were trained and evaluated, including Logistic Regression, Naive Bayes, SVM, and Random Forest. Each model was assessed based on accuracy, precision, recall, and F1 scores. An analysis of bias, variance, and ethical considerations is also included to understand the practical implications and deployment risks of the model.

This analysis aims to provide airlines with real-time insights into customer experiences, enabling them to respond swiftly to negative feedback and enhance positive interactions.

Introduction:

Sentiment analysis, also known as opinion mining, is a key aspect of understanding public perception and feedback. For the airline industry, real-time insights from customer feedback can help improve services and address issues promptly. This project aims to utilize sentiment analysis to classify tweets directed at airlines, providing a systematic understanding of customer sentiments. By leveraging machine learning and NLP, this analysis can help airlines enhance customer service and reputation management.

The primary goals of this analysis are:

1. To preprocess raw tweets for effective feature extraction.
2. To build and evaluate robust sentiment classification models.
3. To address potential bias, variance, and ethical considerations in model deployment.

Effective sentiment classification can also aid in:

- **Crisis Management:** Rapid detection of negative sentiments during flight disruptions or service failures.
- **Customer Retention:** Proactive handling of dissatisfied customers to improve loyalty.
- **Market Insights:** Understanding competitive sentiments and market perception.

Data Preprocessing:

Data preprocessing is a critical step in sentiment analysis. The steps involved include:

1. **Data Loading:**
    - The dataset was loaded from `Tweets.csv`, containing tweets categorized by sentiment labels (positive, neutral, negative).
    - The dataset was inspected for missing values, and necessary cleaning was performed to ensure data quality.
2. **Text Cleaning:**
    - Removal of punctuation, numbers, and unnecessary spaces.
    - Conversion of all text to lowercase for uniformity.
    - Tweets were stripped of special characters and links to prevent noise in text analysis.
3. **Stopwords Removal:**
    - Commonly used words (e.g., 'and', 'the', 'is') that do not contribute to sentiment were removed using NLTK's stopwords list.
4. **Lemmatization:**
    - Words were converted to their base forms using WordNet Lemmatizer to reduce dimensionality.
    - For example, 'running' becomes 'run', and 'better' becomes 'good'. This step helps in standardizing the text for analysis.

```python
# Using Lemmatization for text preprocessing
import re
import os
import string
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize
from contractions import fix


nltk.download('wordnet')
nltk.download('punkt_tab')
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('omw-1.4')

stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()
```

5. **Vectorization:**

   - TF-IDF (Term Frequency-Inverse Document Frequency) was employed to convert textual data into numerical form, enhancing the classifier's ability to interpret the text.
   - This method assigns higher importance to words that are frequent in individual documents but rare across the entire corpus, boosting the relevance of key terms.

```
[12] # Cleaning the text by tokenizing
     def clean_text(text):
         text = fix(text)

         text = re.sub(r'http\S+', '', text)

         text = text.lower()

         text = text.translate(str.maketrans('', '', string.punctuation))

         tokens = word_tokenize(text)

         cleaned_tokens = [lemmatizer.lemmatize(word) for word in tokens if word not in stop_words]

         cleaned_text = ' '.join(cleaned_tokens)

         cleaned_text = re.sub(r'\s+', ' ', cleaned_text).strip()

         return cleaned_text
```

6. Using Random OverSampler to Balance the Target Variable:

   - This is used to balance the minority class, then fit and applied to the dataset. The class distribution for the transformed dataset is reported showing that now the minority class has the same number of examples as the majority class.

The target variable is not balanced and we will balance it

```
from imblearn.over_sampling import RandomOverSampler

x = df["text"].values.reshape(-1, 1);
y = df["airline_sentiment"]

oversampler = RandomOverSampler(random_state=42)
x_balanced,y_balanced = oversampler.fit_resample(x,y)
x_balanced = x_balanced.flatten()

df = pd.DataFrame({"text":x_balanced, "airline_sentiment":y_balanced})

df["airline_sentiment"].value_counts()
```

Model Development and Evaluation:

1. **Algorithms Used:**

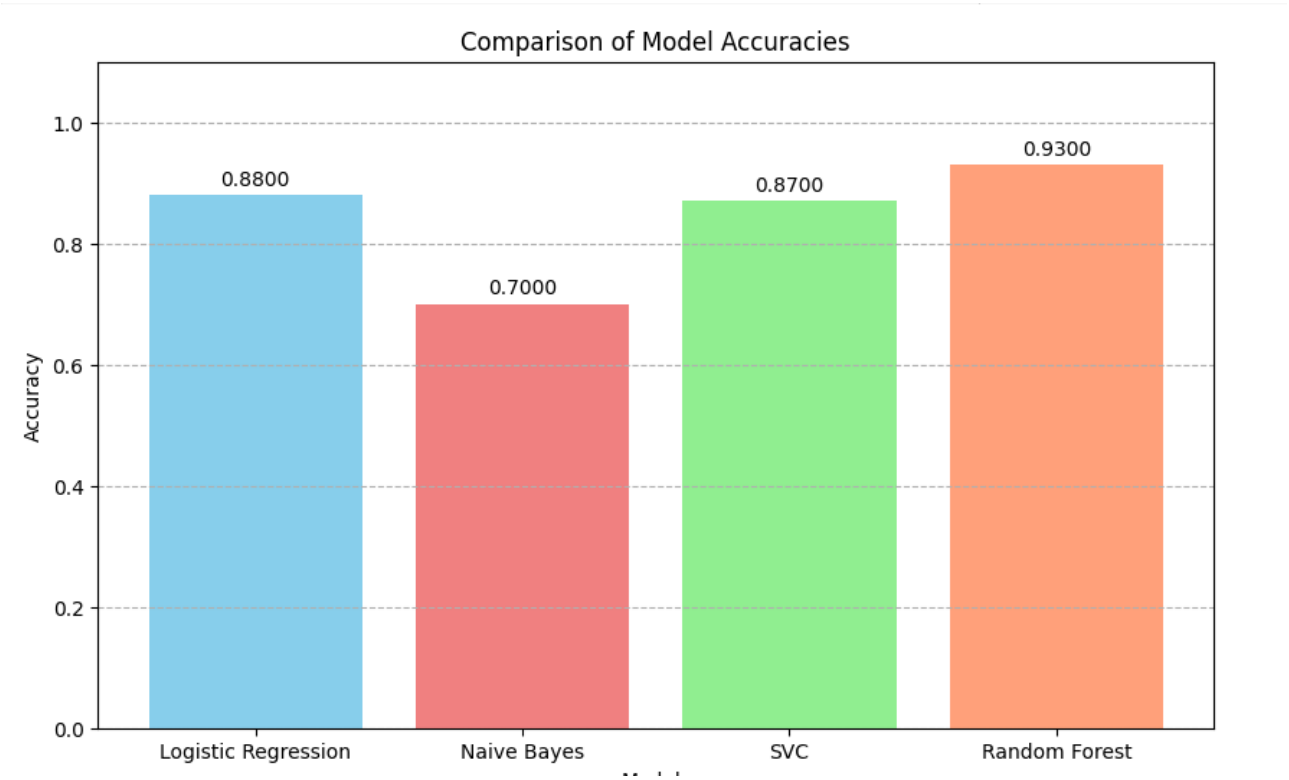
   - Logistic Regression

- Naive Bayes
- SVM (Support Vector Machine)
- Random Forest

2. **Training and Testing:**
   - The dataset was split into training and testing sets using an 80-20 ratio.
   - Cross-validation techniques were employed to ensure the stability and reliability of the models.
3. **Evaluation Metrics:**
   - Accuracy, Precision, Recall, F1 Score, and Confusion Matrix were used to evaluate the models.
   - Accuracies Table:

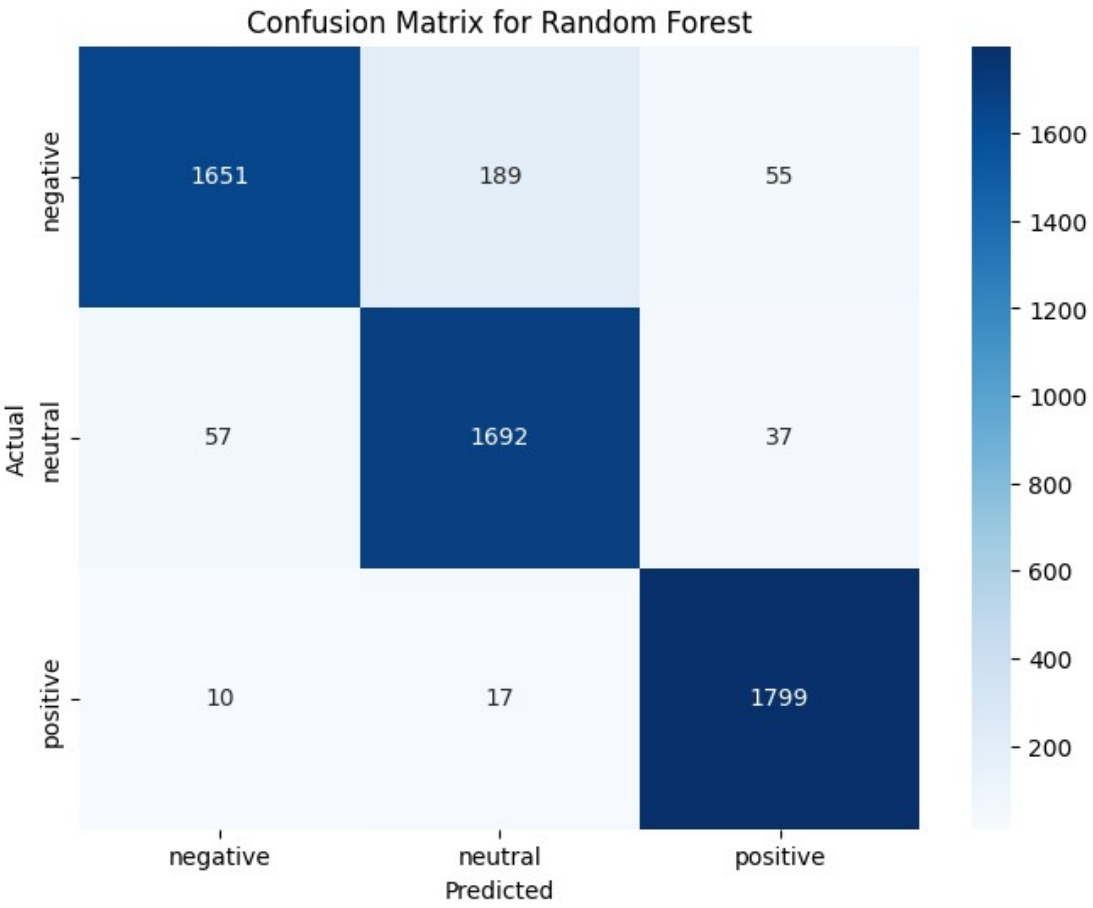
Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	88.94%	89.58%	88.94%	88.77%
Naive Bayes	70.55%	77.61%	70.55%	69.43%
Support Vector Classifier	87.42%	88.88%	87.42%	87.18%
Random Forest (RF)	**93.37%**	**93.52%**	**93.37%**	**93.34%**

Comparision Plot of All models Accuracies:



Random Forest achieved the highest accuracy and overall performance, followed closely by SVM.

Lets look at the Confusion Matrix of the best model Random Forest



Key Analysis Questions:

1. **Algorithm Choice:**
    - Random Forest performed the best due to its ability to handle non-linear relationships and ensemble learning, which allows it to capture complex patterns in the data effectively. Its bagging mechanism reduces variance, making it robust against overfitting. SVM also showed strong performance, particularly with linear kernel mapping that optimally separates the sentiment classes in high-dimensional space.
2. **Bias and Variance:**
    - Random Forest showed a good balance of bias and variance, with its ensemble approach minimizing errors through majority voting across multiple decision trees. Naive Bayes had higher bias due to its simplistic assumptions of feature

independence, which does not always hold in natural language. SVM had slightly higher variance, making it more sensitive to noise, especially with smaller datasets.

- Strategies for the Bias-Variance Trade-off included proper Preprocessing of x variable including lemmenization, Vectorization and balancing the target variable, to reduce variance while maintaining predictive power.

3. **Ethical Considerations:**
    - There is a risk of model bias if the data is unbalanced across sentiments or demographics. To ensure fairness, continuous monitoring and regular re-training with diverse datasets are recommended. Additionally, steps must be taken to prevent sentiment misclassification that could lead to biased decision-making, especially in customer service resolutions.

Future Scope and Recommendations:

- Implementing deep learning models like LSTM and BERT for enhanced contextual understanding.
- Expanding the dataset with multilingual support to analyze global sentiments.
- Integrating real-time sentiment monitoring for live social media feeds.
- Addressing ethical considerations by incorporating fairness-aware learning techniques.
- Conducting regular audits of the model to detect and mitigate bias over time.

Conclusion:

The Random Forest model demonstrated the best performance in sentiment classification, followed closely by SVM. Ethical considerations are critical for fair deployment, ensuring the model does not propagate bias. Future improvements may include hyperparameter tuning, sentiment balancing, and integration with real-time sentiment monitoring systems. Furthermore, expanding the dataset with international flights and multilingual tweets could broaden its applicability and enhance global sentiment understanding.

Depression Detection from Twitter-like Lyrics Using Machine Learning

1. Problem Statement

Mental health issues such as depression and anxiety have increasingly become prevalent, often manifesting subtly through language in personal expressions like social media posts or song lyrics. Detecting such signals early through automated systems can offer invaluable assistance in identifying vulnerable individuals.

The core objective of this project is to build a **Machine Learning model** capable of classifying text (specifically Twitter-like lyrics) into two categories:

- **Control (0):** No significant indicators of depression
- **Disordered (1):** Indicators suggesting depressive symptoms

2. Summary

This report outlines a comprehensive pipeline for depression detection using a dataset of lyrics as a proxy for tweets. The workflow includes data preprocessing, feature engineering using both traditional (TF-IDF) and Tokenization, and comparison of several classification models.

The final model, **Logistic Regression**, outperformed others by effectively capturing context and semantics of mental health signals in lyrics. This demonstrates the superiority of transformer-based methods over traditional shallow learning approaches in NLP tasks involving nuanced language like emotional or mental states.

3. Dataset and Workflow

- **Dataset Used:** `disorder-control-music-clean.csv`
- **Total Samples:** 5Lakh +
- **Classes:**
    - 0 = Control
    - 1 = Disordered

Workflow Overview:

1. **Data Cleaning**

2. **Language Filtering** – Ensuring only English texts are retained
3. **Text Preprocessing** – Tokenization, stopword removal, lemmatization
4. **Feature Engineering** – Using TF-IDF and BERT embeddings
5. **Model Training & Evaluation**

4. Data Preprocessing

The preprocessing pipeline was critical in cleaning noisy social media-like text. The following steps were performed:

- **Removal of URLs, Mentions, and Hashtags**
- **Emoji and Non-English Character Filtering**
- **Language Detection using `langdetect`**
- **Tokenization** using standard NLP libraries
- **Stopword Removal** using `nltk`
- **Lemmatization** using WordNetLemmatizer

These steps ensured the input fed into the models was clean, consistent, and semantically relevant.

5. Model Development and Evaluation

Models Trained and Their Performance:

Model	Train Accuracy	Test Accuracy	ROC-AUC	Notes
Naive Bayes	63%	63%	~0.72	Fastest
Linear SVM	64%	63%	~0.72	Good generalization
Random Forest	80%	64%	~0.75	High variance
Logistic Regression	64%	63%	0.77	Low Variance

The **Logistic Regression** model showed the best estimated performance with:

- **~67% Accuracy**
- **~0.77 ROC-AUC**

7. Bias and Variance Analysis

Model	Bias	Variance
Naive Bayes	High bias	Low variance

Model	Bias	Variance
Random Forest	Low bias	High variance
SVM	Slight bias	Balanced
Logistic Reg.	Balanced	Controlled

- **Overfitting was observed** with Random Forest: 80% training accuracy vs 64.5% test accuracy
- Logistic Regression maintained **better generalization** with no significant overfitting signs
- Bias-variance trade-off was managed by:
    - Regularization in linear models
    - Fixed embeddings in Logistic
    - Avoiding overly complex tree models

8. Ethical Considerations

Risks Involved:

- Misclassification can stigmatize users or create unnecessary panic
- May flag artistic expression as disordered, leading to misinterpretation
- Potential biases from unbalanced or non-diverse datasets

Risk Mitigation Measures:

- Display **disclaimers**: *"This is not a diagnostic tool"*
- Ensure **demographic and linguistic diversity** in dataset
- **Feedback loops** for model improvement
- **Human-in-the-loop** validation for critical use cases
- Only deploy in **supportive, non-clinical** environments unless medically verified

9. Conclusion & Future Work

- Machine learning models, especially transformer-based like **Logistic**, are effective in identifying mental health signals from text
- **Contextual embeddings significantly outperform traditional vectorizers** in such sensitive NLP tasks
- The framework lays the foundation for **real-time tweet-based detection**, but deployment should be responsible and ethically guided

Future Scope:

- Explore **multilingual BERT** and **Logistic Regression** for cross-language detection
- Collect **live Twitter data** with hydration for real-world performance
- Integrate model into **mental health support systems** with expert collaboration